Measuring and Explaining Geomorphic Change in the McMurdo Dry Valleys, Antarctica

Summary. The McMurdo Dry Valleys hold the only streams in Antarctica that flow over open ground. They flow for a few weeks each summer, and they are changing. A 2026 dissertation (Barlow) mapped two decades of that change but stopped short of explaining it — and measured it only inside the stream channels. This project proposes to (1) identify which climate drivers control the change, (2) measure geomorphic change across the whole valley-floor landscape, not just in the channels, and sort it by the process responsible, and (3) attach an honest, per-pixel error bar to every change map. This document is a plan. No results are claimed here; the sections below state plainly what exists, what does not, what the work needs, and what could go wrong.

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1. Background

The McMurdo Dry Valleys (MDVs) are the largest ice-free region of Antarctica. For most of a century they were treated as the most stable landscape on Earth. That view is now in question. The valleys are energy-limited: frozen water is everywhere, but there is barely enough summer heat to melt any of it. A small warming therefore produces a direct, measurable response — ephemeral streams (channels that flow only during the brief melt season) carry more water, move more sediment, and reshape their own beds.

What already exists. Barlow (2026) built the measurement foundation:

- A neural network (U-Net) that finds stream channels from terrain shape alone —

elevation, slope, aspect — with no need for water or color in the data. This matters because the channels are dry 11 months of the year.

- Evidence that the free, continent-wide REMA satellite elevation model is accurate

enough to detect stream-corridor change, once aligned to airborne lidar.

- Maps of elevation change (erosion and deposition) across four valleys and three survey

epochs: 2001 airborne lidar, 2014 airborne lidar, and 2021–23 satellite data.

What does not exist. The dissertation is descriptive, and it stops at the channel edge. It shows *where* in-channel change happened. It does not test *why* — which climate variables drive the change (named in the dissertation as future work) — and its analysis is masked to the stream channels, so everything the elevation differencing measures *outside* them (thawing ground, slopes, fans, lake margins) was discarded unexamined. This proposal is that unfinished work.

Relevance to NASA. The 2001 baseline is NASA data — an ATM airborne lidar campaign — that has never been fully exploited for surface change. This project converts that NASA archive, together with free satellite elevation models and reanalysis, into a continuing, uncertainty-calibrated record of climate-driven surface change in the Dry Valleys: a direct contribution to the Earth Science Division's cryosphere and Earth-surface-change focus areas. The methods deliverables travel beyond the study site — terrain-only feature detection and per-pixel change thresholds apply to any elevation-differencing problem, including the surface-change records NASA missions such as ICESat-2 and NISAR produce. The Dry Valleys are also the canonical terrestrial analog for cold-desert planetary surfaces, which is why NASA has repeatedly funded work there. Finally, the project develops a Future Investigator in exactly the skills — open geospatial data, machine learning on remote sensing, honest uncertainty — that the Division's open-science strategy calls for.

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2. Objectives

O1 — Attribution. Determine which climate drivers control each stream's rate of geomorphic change. Candidate drivers: air temperature (summed as positive degree-days, a running total of melting weather), incoming sunlight, meltwater discharge, glacier mass balance, and summer soil-thaw depth. *Hypothesis: cumulative melt energy predicts sediment movement better than water volume alone.* This is testable and falsifiable — if discharge alone predicts just as well, the hypothesis fails and that is a publishable answer too.

O2 — Landscape-wide geomorphic change. Extend the change measurement from inside the stream channels to the whole valley-floor surface, and classify each patch of significant change by the process responsible: channel shift, thaw-driven subsidence (thermokarst), slope movement, fan growth, lake-margin change. *Hypothesis: different process types answer to different drivers — channels to melt energy and water, thermokarst to summer thaw depth.* If the types turn out to share one driver, that is also an answer.

One practical note on satellite data: the record only continues past 2014 on the free REMA satellite DEM, and satellite elevations disagree with lidar in terrain-dependent ways. Measuring and correcting that disagreement is part of the method (§3), not an objective.

O3 — Calibrated uncertainty. Replace the single valley-wide noise estimate used today with a per-pixel detection threshold that accounts for slope, aspect, and sensor. The test of success is strict: on ground that did not change, a 95%-confidence threshold should be exceeded by roughly 5% of pixels — no more, no less.

Together the three objectives turn a one-time survey into a continuing, self-checking monitoring capability with an explanatory model behind it.

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3. Planned approach

O1 (Years 1–2). 1. Compute each gauged stream's change rate: difference two elevation epochs (DEM-of-Difference), keep only changes larger than the detection floor, sum inside the stream's channel outline, normalize by channel area and by years elapsed. 2. Build each stream's driver history from the MDV-LTER station records (meteorology, 21 stream gauges, 7 glacier mass-balance series, soil temperature) and from the ERA5 and AMPS weather models where station coverage has gaps. 3. Fit two model families — hierarchical regression (interpretable) and random forest (flexible) — relating rate to drivers. Validate only on streams held out of fitting. Agreement between the two families is the sanity check. 4. Test the melt-energy hypothesis directly against the water-volume alternative.

O2 (Year 2). 1. Apply the O3 per-pixel thresholds to the full valley-floor surface and extract every patch of change that clears them. 2. Classify each patch by process type from its shape and setting (slope position, distance to channels and lakes, ice-cored terrain). 3. Run each type's rates through the O1 driver models separately and compare the driver fingerprints.

Sensor continuity (method, Years 1–2). Quantify satellite-vs-lidar disagreement over stable ground by slope and aspect; correct it before any differencing that mixes sensors. Extend beyond Taylor Valley where REMA quality allows.

O3 (Years 1–3, alongside). 1. Model the measurement noise as a function of slope, aspect, and sensor instead of one global number. 2. Publish the result as a per-pixel threshold map and verify the 5%-on-stable-ground property on held-out terrain.

Methods note. All elevation differencing uses robust statistics (NMAD, an outlier-resistant standard deviation) because elevation errors in this terrain are known to be heavy-tailed; a naive standard deviation would be inflated by a handful of blunders.

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4. What the project needs

Data. Everything required is free and public except one item.

Need	Source	Availability
2001 airborne lidar DEM (2 m)	NASA ATM via USGS ScienceBase	public, free
2014 airborne lidar DEM (1 m) + point cloud	NCALM via OpenTopography	public, free
REMA satellite DEM (2 m, 2021–23)	Polar Geospatial Center / AWS	public, free
Stream discharge, 21 gauges (daily)	MCM-LTER via EDI	public, free
Meteorology, glacier mass balance, soil temperature	MCM-LTER via EDI	public, free
ERA5 reanalysis; AMPS Antarctic weather model	Copernicus CDS; NCAR GDEX	public, free
Stream channel outlines (labels)	MCM-LTER via EDI	public, free
Barlow's hand-digitized training tiles	dissertation author	author-gated — must be requested

Labels. The detector needs training examples. The public LTER channel outlines are a workable starting set. The gold-standard set — the dissertation author's hand-digitized tiles — is not published and must be requested directly. If that request fails, a comparable set will be re-digitized by hand. This costs weeks, not months, and is budgeted in Year 1.

Computing. A single workstation GPU is sufficient for training the segmentation models; the elevation processing is CPU-bound and modest. No supercomputer allocation is required. Full-valley raster stacks run to tens of gigabytes and need local storage, not specialized infrastructure.

Skills and mentoring. The FI has built terrain-segmentation and change-detection pipelines on non-Antarctic data but is new to Antarctic hydrology and to hierarchical statistical modeling — a genuine gap, stated as one. How coursework and PI mentorship close it is detailed in the Research Readiness Statement and Mentoring Plan.

Community contact. The dissertation author has been contacted and is responsive. Continued cooperation is helpful but not load-bearing: every analysis in this plan can proceed from public data alone.

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5. Honest assessment

What could go wrong, and what happens then.

1. The attribution signal may be weak. With ~20 gauged streams and three epochs, the sample is small. If no driver clearly outperforms the others, the deliverable becomes a rigorous null result with calibrated uncertainty — less exciting, still useful, and still publishable. The risk is to impact, not to feasibility. 2. Gauge records are patchy after 2015. Stream gauging in the MDVs declined; some streams have only a few dozen recorded days in the satellite epoch. This is precisely why O1 leans on modeled melt energy (which is continuous everywhere) rather than gauges alone — but it means the discharge side of the hypothesis test is weakest in the most recent period, and the proposal accepts that. 3. Satellite coverage is uneven. REMA quality varies by location and date. Taylor Valley has the best three-epoch coverage and will anchor any acceleration claims; other valleys may support only two epochs. Conclusions will be scoped accordingly. 4. The label request may be declined. Mitigation is above (re-digitize). This adds Year-1 labor but does not block the science. 5. Outside the channels, change may be too small to detect (O2). The valley floors beyond the streams change slowly; much of the surface may sit below the detection threshold even with per-pixel calibration. If so, O2 narrows back to in-channel change and the landscape-wide result is reported as a calibrated null — which is itself a quantitative statement about how spatially confined active change currently is.

What this proposal does not claim. No preliminary results are presented. The dissertation being extended is the prior work; the FI's contribution begins at award. The hypotheses may be wrong. The plan is designed so that every objective produces a defensible, publishable answer whether its hypothesis survives or not.

Why it is worth doing anyway. The MDV streams are the most direct, physically measurable climate response in the most stable landscape on Earth, and the data to study them — two lidar epochs, a growing satellite record, and 30 years of LTER station records — already exist and are free. The missing ingredients are the driver analysis, the whole-landscape picture, and the error bars. Those are exactly the size of a three-year graduate project.

6. Timeline and deliverables

Period	**Work**	**Deliverable**
Yr 1	Rebuild the change-detection chain from public data; harmonize driver records per stream; secure or rebuild training labels; first attribution model (Taylor Valley)	Attribution manuscript drafted; labeled training set archived
Yr 2	Landscape-wide change detection + process-type classification (O2), gated by the O3 thresholds; driver tests per change type; sensor-disagreement correction	Process-classification paper; AGU presentation
Yr 3	Per-pixel calibrated-uncertainty product; three-epoch acceleration analysis; extension beyond Taylor Valley where REMA supports it; synthesis	Synthesis paper; public data products with DOIs

7. Data management and open science

All inputs are public and cited to source (NASA ATM, NCALM/OpenTopography, PGC REMA, MCM-LTER/EDI, Copernicus ERA5, AMPS/GDEX). All derived products — change rasters, process-classified change maps, per-stream rate tables, uncertainty layers, channel outlines, and processing code — will be released open-access with DOIs (Zenodo / EDI). Heavy regenerable rasters stay out of version control; scripts, tables, and figures are tracked. Products carry explicit coordinate-system, method, and uncertainty metadata.

References

[1] Barlow, M. C. (2026). *A Comprehensive Spatial Analysis of Stream Boundary and Geomorphological Change Detection: McMurdo Dry Valleys, Antarctica.* PhD Dissertation, University of Houston (chair: C. L. Glennie).

[2] Barlow, M. C., Zhu, X., & Glennie, C. L. (2022). Stream Boundary Detection of a Hyper-Arid, Polar Region Using a U-Net Architecture: Taylor Valley, Antarctica. *Remote Sensing* 14(1):234.

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[4] Howat, I. M., et al. (2019). The Reference Elevation Model of Antarctica (REMA). *The Cryosphere* 13:665–674.

[5] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI* 234–241.

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[7] Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146:1999–2049.

Plan-only draft ("basic" variant): no preliminary results, no figures. Companion variants of the same plan: `finesst_proposal.md` *(formal) and* `finesst_proposal_plain.md` *(plain language). Not a submitted proposal.*